contiguous PAFTP packet sequence numbers are already received. If this timer is not expired, the peer MAY send a pending acknowledgement as a stand-alone PAFTP packet, and acknowledgement number SHALL be the PAFTP packet sequence number of which all previous contiguous PAFTP packet sequence numbers are already received.

Because the server's handshake response bears an implicit PAFTP sequence number of zero, a client SHALL set its pending acknowledgement value to zero and start its send-acknowledgement timer when it receives the server's a handshake response. Operation of the send-acknowledgement and acknowledgement-received timers is illustrated in Figure 33, "PAFTP session lifecycle for Commissioner" in Section 4.20.3.11, "Protocol State Diagrams".

If a peer detects that its receive window has shrunk to two or fewer free slots, it SHALL immediately send any pending acknowledgement as a stand-alone PAFTP packet, and acknowledgement number SHALL be the PAFTP packet sequence number of which all previous contiguous PAFTP packet sequence numbers are already received. This prevents the session from stalling in the interval between when a peer's receive window becomes empty and when its send-acknowledgement timer would normally fire.

4.20.3.9. Idle Connection State

When neither side of a PAFTP session has data to send, PAFTP packets will still be exchanged every send-acknowledgement interval due to acknowledgements generated by the receipt of previous data or stand-alone acknowledgement packets, as discussed in Section 4.20.3.8, "Packet Acknowledgements". The behavior of the acknowledgement-received timer in this scenario doubles as a keep-alive mechanism, as it will cause a peer to close a WFA-USD connection automatically if the remote PAFTP stack crashes or becomes unresponsive. This scenario is illustrated in Figure 32, "Idle connection scenario".

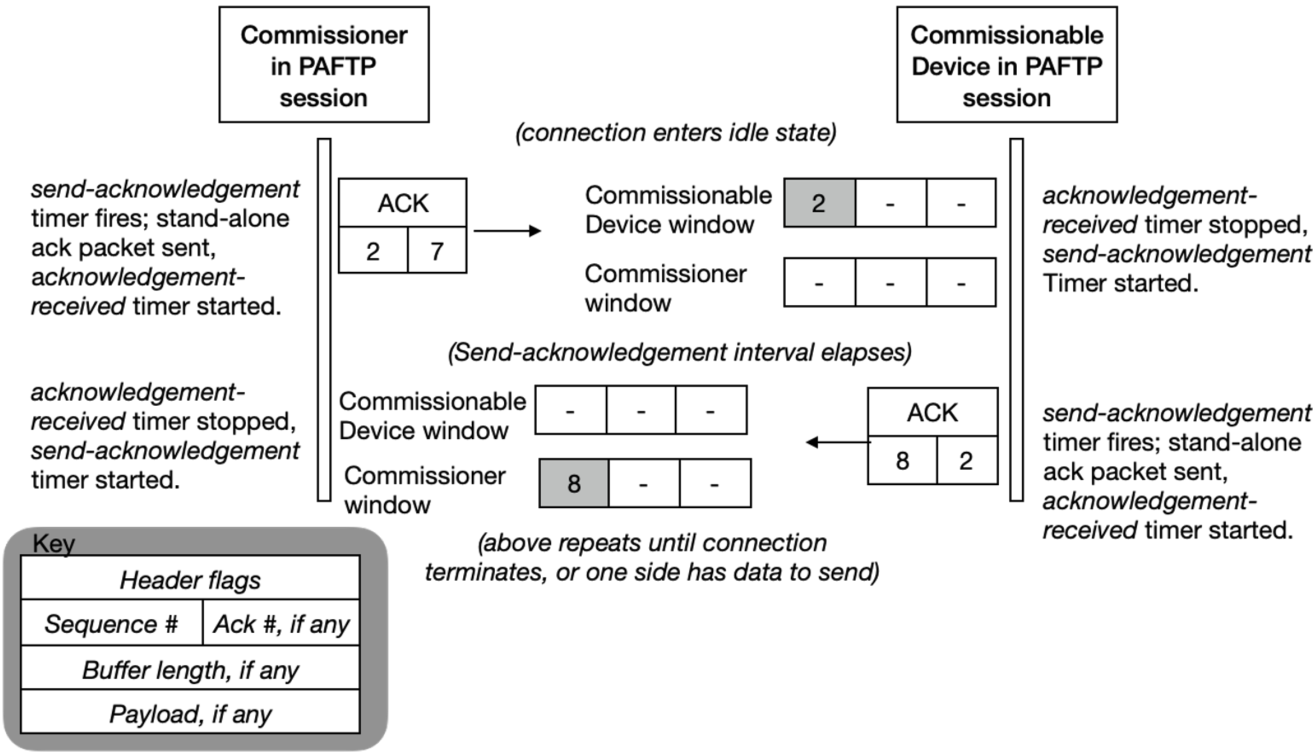

Figure 32. Idle connection scenario